Fourier covariance and positivity on the three-torus

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Abstract

For normalized even Fourier symbols on the three-torus, we give complete proofs connecting anchored covariance, conditional negativity, pointwise iterated Gamma forms, finite heat-multiplier matrices and positivity on the full nonnegative Haar-L^2 cone. The passage from finite Fourier squares to arbitrary nonnegative L^2 functions is proved explicitly. Coordinatewise power symbols have the sharp positivity condition that every positive-weight exponent is at most two; inactive coordinates impose no restriction. We also construct the unique actual torus probability laws, their convolution kernels and countable-step chains. A genuine Poisson mixture has the exact transformed symbol, a persistent no-jump atom and a short-parameter jump-covariance identity. We identify the exact closed generator domain and explain why semigroup existence alone supplies no positivity. A bounded rank-two multiplier has an explicit nonnegative test sent outside the Haar cone at every positive time. Exact integrated Gamma dissipation and sharp all-level decay are also proved, including non-Markov power orders. A finite-time increment kernel remains positive at every time, yet an exact two-frequency quadratic-heat example shows that it need not increase in Loewner order. The main equivalence is classical Schoenberg theory, organized here around a frozen Lean development. No new priority claim, completed pointwise Gamma domain or continuous-time path construction is asserted.

1 The question and the domains

Let $\mathbb{T}^3 = (\mathbb{R}/\mathbb{Z})^3$, with normalized Haar measure m, and put $\Lambda = \mathbb{Z}^3$. For $k \in \Lambda$ write

$$e_k(x) = \exp(2\pi i k \cdot x).$$

The finite Fourier algebra $\mathcal{P}$ consists of the finite sums $f = \sum_k \widehat{f}(k) e_k$. Its product is pointwise multiplication, $e_k e_l = e_{k+l}$, and complex conjugation sends e_k to e_{-k}. The exponentials form an orthonormal basis of $L^2(\mathbb{T}^3, m; \mathbb{C})$. We use the unit-period normalization throughout. In particular the rates below are specified on integer frequencies; a factor 2π is not silently inserted into their definition.

For a real symbol $\psi : \Lambda \to \mathbb{R}$, the algebraic operator L_ψ is defined on $\mathcal{P}$ by

$$L_\psi e_k = -\psi(k) e_k. \tag{1}$$

Until a closed domain is specified, this is only a finite Fourier operator. The sesquilinear iterates used in the formal library are

$$\begin{aligned} \Gamma_0(f,g) &= \overline{f} g, \\ \Gamma_{n+1}(f,g) &= \tfrac{1}{2}\big(L_\psi \Gamma_n(f,g) - \Gamma_n(L_\psi f, g) - \Gamma_n(f, L_\psi g)\big). \end{aligned} \tag{2}$$

They are conjugate linear in f and linear in g. For an even real symbol, $L_\psi \overline{f} = \overline{L_\psi f}$, and Γ_1 is the usual complex carré du champ. We keep the recursion itself as the definition when discussing a noneven symbol in Section 4.

Define the anchored kernel

$$K_\psi(k,l) = \frac{\psi(k) + \psi(l) - \psi(l-k)}{2}. \tag{3}$$

Saying that K_ψ is positive semidefinite means that every finite matrix $(K_\psi(k_i,k_j))_{i,j=1}^n$ is real symmetric and has nonnegative quadratic form. Repetitions of the sampled frequencies are allowed. Conditional negativity means that

$$\sum_{i,j=1}^{n} c_i c_j \psi(k_j - k_i) \le 0 \quad \text{whenever } c_i \in \mathbb{R}, \quad \sum_i c_i = 0. \tag{4}$$

Symmetry is stated separately; condition (4) by itself does not test the odd part of a real symbol.

The physical nonnegative cone in complex L^2 is

$$\mathcal{C} = \{u \in L^2(\mathbb{T}^3, m; \mathbb{C}) : \operatorname{Im} u = 0,\ \operatorname{Re} u \ge 0 \text{ almost everywhere}\}.$$

Thus an inequality about Fourier coefficients is not the definition of positivity used here.

Theorem 1.1. *Suppose $\psi : \Lambda \to [0,\infty)$ is even and $\psi(0) = 0$. Let S_t on L^2 be the Fourier multiplier $\widehat{S_t u}(k) = e^{-t\psi(k)}\widehat{u}(k)$, for $t \ge 0$. The following conditions are equivalent:*

1. *K_ψ is positive semidefinite on every finite sample;*
2. *ψ satisfies (4);*
3. *$\Gamma_1(f,f)(0) \ge 0$ for every $f \in \mathcal{P}$;*
4. *$(e^{-t\psi(k_j-k_i)})_{i,j}$ is positive semidefinite for every finite sample and every $t \ge 0$;*
5. *$S_t\mathcal{C} \subseteq \mathcal{C}$ for every $t \ge 0$.*

Under these conditions, $\Gamma_n(f,f)(x)$ is real and nonnegative for every integer $n \ge 0$, $f \in \mathcal{P}$ and $x \in \mathbb{T}^3$. The semigroup is strongly continuous, self-adjoint, contractive and constant preserving. Its exact generator domain is

$$\operatorname{Dom}(A) = \left\{ u \in L^2 : \sum_k \psi(k)^2 |\widehat{u}(k)|^2 < \infty \right\}, \qquad \widehat{Au}(k) = -\psi(k)\widehat{u}(k). \tag{5}$$

Finite Fourier polynomials form a core for this closed generator.

The finite equivalence in this statement is Schoenberg's classical mechanism [2]. Its value here is the complete connection to the specified physical L^2 realization, including the square-cone closure and the exact operator domain. All proofs are given below. Positivity of each finite matrix is not used as a substitute for proving positivity on $\mathcal{C}$.

2 Anchoring a conditionally negative kernel

The zero frequency is needed as an extra sample even when it does not occur in the polynomial of interest.

Lemma 2.1. *Let ψ be real, even and normalized by $\psi(0) = 0$. Then K_ψ is positive semidefinite on finite samples if and only if* (4) *holds.*

Proof. Evenness makes K_ψ symmetric. For any coefficients b_i, write $b = \sum_i b_i$. Expanding (3) gives

$$\sum_{i,j} b_i b_j K_\psi(k_i, k_j) = b \sum_i b_i \psi(k_i) - \frac{1}{2} \sum_{i,j} b_i b_j \psi(k_j - k_i). \tag{6}$$

For $b = 0$, positivity of the left side yields conditional negativity.

Conversely, augment the sample by $k_0 = 0$ and give it coefficient $b_0 = -\sum_{i=1}^n b_i$. The augmented coefficients sum to zero. The diagonal term at the anchor vanishes, and its two cross terms are $-2b \sum_i b_i \psi(k_i)$, by evenness. Therefore the conditionally negative quadratic form of the augmented sample is exactly -2 times the left side of (6). Its nonpositivity proves the desired matrix positivity for arbitrary b_i. □

A real positive semidefinite matrix M also has nonnegative complex energy: if $z = a + ib$, symmetry gives

$$\sum_{i,j} \overline{z}_i z_j M_{ij} = a^{\mathsf{T}} M a + b^{\mathsf{T}} M b \geq 0.$$

This observation permits complex Fourier polynomials in all subsequent pointwise tests without changing the real definition of conditional negativity.

3 The entire hierarchy on the finite algebra

For a single pair of modes, recursion (2) gives

$$\Gamma_n(e_k, e_l) = K_\psi(k, l)^n e_{l-k}. \tag{7}$$

Indeed the formula is true for $n = 0$; the next recursion multiplies its coefficient by $(-\psi(l-k) + \psi(k) + \psi(l))/2$. The symbol is real, so conjugating the coefficient $-\psi(k)$ in the first argument does not change it. Finite sesquilinearity now yields

$$\Gamma_n(f, f)(x) = \sum_{i,j} \overline{c_i e_{k_i}(x)} c_j e_{k_j}(x) K_\psi(k_i, k_j)^n, \quad f = \sum_i c_i e_{k_i}. \tag{8}$$

Here $M^{\circ n}$ denotes the entrywise power, not the matrix power M^n.

Lemma 3.1. *If M and N are real positive semidefinite finite matrices, then their entrywise product $M \circ N$ is positive semidefinite. Consequently $M^{\circ n}$ is positive semidefinite for every integer $n \geq 0$.*

Proof. Choose finite-dimensional Gram representations $M_{ij} = \langle u_i, u_j \rangle$ and $N_{ij} = \langle v_i, v_j \rangle$, for example by orthogonal diagonalization and square roots of their nonnegative eigenvalues. Then $(M \circ N)_{ij}$ is the Gram matrix of $u_i \otimes v_i$ in the finite tensor-product Euclidean space. Iteration proves the power assertion, starting from the all-ones Gram matrix $M^{\circ 0}$. □

Proposition 3.2. *For a real even normalized symbol, the following are equivalent: K_ψ is positive semidefinite; $\Gamma_1(f,f)(0) \geq 0$ for every finite Fourier polynomial; and all the values $\Gamma_n(f,f)(x)$ are real and nonnegative.*

Proof. If the kernel is positive semidefinite, apply Lemma 3.1 to (8), using the complex energy observation after Lemma 2.1. This proves the full hierarchy. The hierarchy implies its first origin test. For the reverse implication, test $f = \sum_i c_i e_{k_i}$ at $x = 0$ with arbitrary real c_i. Since $e_{k_i}(0) = 1$, equation (8) at $n = 1$ is exactly the quadratic form of K_ψ on that sample. Repeated modes cause no difficulty: the formula is valid for every finite presentation of f. □

This result is positivity on $\mathcal{P}$, not an assertion that the products in (2) extend to all of $\mathrm{Dom}(A)$. The exact generator domain alone does not control the products or pointwise representatives required there.

4 Normalization and real-symbol pitfalls

The evenness assumption is structural. It must not be dropped merely because a scalar quadratic form is called negative.

Proposition 4.1. *Let $\psi : \Lambda \to \mathbb{R}$ satisfy $\psi(0) = 0$, without initially assuming evenness. If $\mathrm{Re}\,\Gamma_1(f,f)(0) \geq 0$ for every $f \in \mathcal{P}$, then ψ is even. It follows that the normalized first-origin-test criterion is equivalent to conditional negativity together with evenness.*

Proof. Fix k and test $f = s + e_k$ with $s \in \mathbb{R}$. Formula (8), which follows from the recursion even without evenness, gives

$$\mathrm{Re}\,\Gamma_1(s+e_k, s+e_k)(0) = \psi(k) + \frac{s}{2}(\psi(k) - \psi(-k)).$$

An affine real function nonnegative for every $s \in \mathbb{R}$ has zero slope. Hence $\psi(k) = \psi(-k)$. Apply Proposition 3.2 and Lemma 2.1 to finish both directions of the criterion. □

For an explicit failure, take $\psi(k) = k_1$. For every real finite vector c, the matrix $\psi(k_j - k_i)$ is antisymmetric, so its quadratic form is zero. In particular it satisfies the bare inequality (4). Yet for $k = (1,0,0)$ and $s = -2$ the displayed first-Gamma value equals $1-2 = -1$. This symbol is not nonnegative and does not fall under Theorem 1.1; it isolates the missing symmetry assumption rather than contradicting that theorem.

Without normalization, positivity of only the real part can also miss an imaginary antisymmetric contribution to (8). Testing $f = e_k + ie_l$ gives

$$\mathrm{Im}\,\Gamma_1(f,f)(0) = K_\psi(k,l) - K_\psi(l,k).$$

Thus the library's unnormalized complex criterion keeps reality as well as nonnegativity. The main theorem avoids this ambiguity by stating normalization and evenness at the outset.

5 Finite Schoenberg positivity

We next pass from the infinitesimal symbol to finite-time multiplier matrices. This step is still a finite matrix argument.

Proposition 5.1. *For real even ψ with $\psi(0) = 0$, conditional negativity is equivalent to positivity of all finite matrices*

$$H_t(i,j) = \exp\bigl(-t\psi(k_j - k_i)\bigr), \qquad t \geq 0. \tag{9}$$

Proof. By Lemma 2.1, assume the anchored matrix $K = (K_\psi(k_i, k_j))$ is positive semidefinite. Its entrywise exponential

$$\exp^\circ(2tK) = \sum_{n=0}^{\infty} \frac{(2t)^n}{n!} K^{\circ n}$$

is positive semidefinite. Each partial sum has that property by Lemma 3.1, and the entrywise convergent finite-matrix limit preserves every quadratic-form inequality and symmetry. Set $d_i = e^{-t\psi(k_i)}$. The identity

$$H_t(i,j) = d_i d_j \exp(2tK_{ij})$$

shows that H_t is the congruence of this positive matrix by the real diagonal matrix with diagonal d_i. Hence it is positive semidefinite.

Conversely, take arbitrary real c_i with $\sum_i c_i = 0$. The finite sum

$$E(t) = \sum_{i,j} c_i c_j e^{-t\psi(k_j - k_i)}$$

is nonnegative for $t \geq 0$ and satisfies $E(0) = 0$. Its right derivative at zero is nonnegative. Differentiating its finite number of terms gives

$$0 \leq E'(0) = -\sum_{i,j} c_i c_j \psi(k_j - k_i),$$

which is conditional negativity. □

No growth condition on ψ is required for this finite argument. For the physical L^2 evolution, nonnegativity of the rates ensures bounded multipliers. Under the main normalized even conditions it also follows from conditional negativity: apply (4) to frequencies $0, k$ and coefficients $1, -1$ to obtain $-2\psi(k) \leq 0$.

6 The completed multiplier and its exact generator

The analytic construction in this section requires only $\psi(k) \geq 0$. Conditional negativity is not needed. Parseval's identity identifies $L^2(\mathbb{T}^3, m; \mathbb{C})$ with $\ell^2(\Lambda; \mathbb{C})$. Since $0 < e^{-t\psi(k)} \leq 1$, its diagonal multiplier defines a contraction S_t. The identities $S_0 = I$ and $S_s S_t = S_{s+t}$ hold coefficientwise. For any fixed u and $t_n \to t$ in $[0, \infty)$,

$$\|S_{t_n} u - S_t u\|_2^2 = \sum_k |e^{-t_n\psi(k)} - e^{-t\psi(k)}|^2 |\widehat{u}(k)|^2 \longrightarrow 0.$$

The summands are bounded by $4|\widehat{u}(k)|^2$; a finite-head, small-tail argument proves the convergence, or equivalently dominated convergence on counting measure. The multiplier is real, so S_t is self-adjoint. If $\psi(0) = 0$, then $S_t 1 = 1$.

Proposition 6.1. *For every nonnegative symbol, the strong right derivative $Au = \lim_{t\downarrow 0}(S_t u - u)/t$ exists exactly on* (5)*. This operator is closed and has $\mathcal{P}$ as a graph core.*

Proof. If the strong derivative exists with value v, each continuous Fourier coordinate functional gives

$$\widehat{v}(k) = \lim_{t\downarrow 0} \frac{e^{-t\psi(k)} - 1}{t} \widehat{u}(k) = -\psi(k)\widehat{u}(k).$$

Parseval then forces $\sum_k \psi(k)^2 |\widehat{u}(k)|^2 < \infty$. Conversely, if this sum is finite, let v have those coefficients. The scalar estimate $0 \le 1 - e^{-a} \le a$ for $a \ge 0$ gives

$$\left| \frac{e^{-t\psi(k)} - 1}{t} + \psi(k) \right| \le \psi(k), \qquad t > 0.$$

The scalar expression tends to zero for every k, so domination by the summable squared sequence $\psi(k)^2|\widehat{u}(k)|^2$ proves convergence to v in L^2.

To prove closedness, suppose $u_n \to u$ and $Au_n \to v$ in L^2. Coordinatewise limits give $\widehat{v}(k) = -\psi(k)\widehat{u}(k)$, and since $v \in L^2$ the same characterization puts $u \in \mathrm{Dom}(A)$ with $Au = v$. Finally truncate the Fourier series of $u \in \mathrm{Dom}(A)$ to $\{k : |k_j| \le N,\ j = 1, 2, 3\}$. Both the L^2 tail and its A-image tail tend to zero by their respective square sums. These truncations are finite Fourier polynomials, proving the core assertion in the norm $\|u\|_2 + \|Au\|_2$. □

The semigroup can therefore exist with a completely controlled closed generator even when it fails positivity. Section 10 gives an explicit failure within this same construction. It cannot be repaired by calling the coefficient evolution a Markov operator before checking the physical cone.

7 A bounded contraction that fails positivity at every positive time

A nonnegative, even, bounded rate can satisfy all elementary Hilbert contraction requirements and still fail the positivity criterion. Set $w = (1, 1, 1) \in \mathbb{Z}^3$ and define

$$\psi(k) = \begin{cases} 1, & k = 2w \text{ or } k = -2w, \\ 0, & \text{otherwise.} \end{cases}$$

Thus $0 \le \psi \le 1$, $\psi(-k) = \psi(k)$ and $\psi(0) = 0$. Let P be the Haar-L^2 orthogonal projection onto $\mathrm{span}\{e_{2w}, e_{-2w}\}$. The actual multiplier is

$$S_t = I + (e^{-t} - 1)P, \qquad A = -P, \qquad \mathrm{Dom}(A) = L^2. \tag{10}$$

On $\operatorname{ran} P$ its multiplier is e^{-t}, and on $\ker P$ it is one. This proves the semigroup law, self-adjointness, contraction and preservation of constants. It also proves operator-norm continuity: $\|S_t - I\| = 1 - e^{-t}$ for $t \ge 0$, and $(S_t - I)/t \to -P$ in operator norm. Hence the entire space is the closed generator domain; no unbounded-domain defect is responsible for the failure below. The nonnegative Hilbert energy is

$$-\langle Au, u\rangle = \|Pu\|_2^2 \ge 0.$$

The two-frequency covariance test is explicit:

$$\big(K_\psi(k,l)\big)_{k,l\in\{w,2w\}} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 1 \end{pmatrix}.$$

Indeed $\psi(w) = 0$, $\psi(2w) = 1$ and $\psi(2w - w) = 0$. For $f = e_w - \frac{1}{2}e_{2w}$, equation (8) gives

$$\Gamma_1(f, f)(0) = -\tfrac{1}{4}. \tag{11}$$

So the rate is neither conditionally negative nor an anchored covariance symbol, despite boundedness, symmetry and nonnegativity.

There is a direct physical cone witness at *every* $t > 0$. Write $\theta = 2\pi w \cdot x$ and take

$$h(x) = (1 - \cos\theta)^2 = \tfrac{3}{2} - 2\cos\theta + \tfrac{1}{2}\cos(2\theta) \geq 0.$$

Only its final harmonic is damped by (10); therefore

$$(S_t h)(0) = \tfrac{1}{2}(e^{-t} - 1) < 0 \qquad (t > 0).$$

This is a finite Fourier polynomial, so its image is continuous and strict negativity at the origin persists on an open neighborhood. Every such neighborhood has positive Haar measure: it contains a small coordinate box in a torus chart, whose product Lebesgue measure is positive. Thus the witness violates the actual almost-everywhere Haar-L^2 cone, not just a choice of pointwise representative.

The bounded rate and the value (11) are mapped in `BoundedRateGammaObstruction`. The whole-space projection formula and the all-positive-time nonnegative test h are explicit human deductions from that rate and the already established physical multiplier. No separate named formal theorem for h is claimed. This is a concrete instance of the classical positivity equivalence, not a new general obstruction principle.

8 Integrated Gamma energy can decay without pointwise positivity

The distinction also survives exact integrated energy identities. Fix $\gamma > 0$, distinct integers $n_1, \ldots, n_N$ and complex coefficients c_j. On the first coordinate circle use

$$\psi_\gamma(k) = |k_1|^\gamma, \qquad f(x) = \sum_{j=1}^{N} c_j e_{n_j e_1}(x), \qquad E_q(f) = \int_{\mathbb{T}} \Gamma_q(f, f)(x_1, 0, 0)\, dx_1.$$

Here q is a nonnegative integer. The zeroth iterate is $|f|^2$. The integral is real, and orthogonality of distinct circle characters in (8) gives the exact formula

$$E_q(f) = \sum_j |n_j|^{\gamma q} |c_j|^2 \geq 0, \tag{12}$$

where the weight is interpreted as one when $q = 0$, including $n_j = 0$. For $q \geq 1$, the zero frequency has zero weight. Indeed the integral eliminates every off-diagonal term and $K_{\psi_\gamma}(ne_1, ne_1) = |n|^\gamma$. This computation needs neither conditional negativity nor positivity of the pointwise Gamma kernel.

Let $f_t = \sum_j e^{-t|n_j|^\gamma} c_j e_{n_j e_1}$. Then

$$E_q(f_t) = \sum_j |n_j|^{\gamma q} e^{-2t|n_j|^\gamma} |c_j|^2, \tag{13}$$

$$\frac{d}{dt} E_q(f_t) = -2E_{q+1}(f_t).$$

The derivative follows by differentiating a finite sum, so no closure or infinite-series interchange has been used. If all frequencies are nonzero, then $|n_j|^\gamma \geq 1$ and

$$E_q(f_t) \leq e^{-2t} E_q(f) \qquad (q \geq 0,\ t \geq 0). \tag{14}$$

The single unit frequency attains equality for every q and t, so the coefficient two in the exponent is sharp at every energy level. For $q \geq 1$ a zero mode is harmless; for $q = 0$ it must be removed to obtain this decay for nonzero constants. Equivalently, one can apply the zeroth-level estimate to the mean-zero part.

These identities hold for every positive γ, even $\gamma > 2$, when the pointwise positivity criterion fails by Section 10. Thus nonnegative circle-integrated energies, exact dissipation and sharp spectral decay do not prove pointwise Gamma positivity, Markov positivity or a probability law. For $0 < \gamma \leq 2$, the positive-law results below apply in addition; for larger orders only the stated multiplier and finite energy assertions are made. No completed pointwise Gamma domain is inferred in either case.

The formal module `StableGammaEnergyDecay` records the actual circle average, diagonal energy, finite semigroup, derivative and sharp unit-frequency witness. Its nonzero-frequency decay statement retains the nonzero-frequency hypothesis even at positive levels; the harmless-zero-mode variant just proved is a labelled elementary human extension of the same finite sum. These are classical Fourier orthogonality and multiplier calculations. They supplement the positivity equivalence rather than supply a new curvature theorem.

9 From finite squares to the full Haar cone

First calculate on a finite polynomial $f = \sum_i c_i e_{k_i}$:

$$(S_t|f|^2)(x) = \sum_{i,j} \overline{c_i e_{k_i}(x)} c_j e_{k_j}(x) e^{-t\psi(k_j - k_i)}. \tag{15}$$

This is equality of continuous finite Fourier representatives. If the matrices (9) are positive semidefinite, the right side is real and nonnegative at every x.

Lemma 9.1. *The L^2 closure of $\{|f|^2 : f \in \mathcal{P}\}$ is exactly $\mathcal{C}$.*

Proof. For $z \in \mathbb{C}$ set $\pi(z) = \max(\operatorname{Re} z, 0)$, regarded as a complex number with zero imaginary part. The map π is one-Lipschitz, vanishes at zero and induces a one-Lipschitz map on L^2. Its fixed-point set is $\mathcal{C}$, hence this cone is closed. Every polynomial square lies in it.

For the reverse inclusion, take $u \in \mathcal{C}$ and choose continuous complex v_n converging to u in L^2. Such approximation follows from continuous-function density for Haar measure on the compact torus. The clipped continuous functions $g_n = \pi(v_n)$ are nonnegative and converge to u, because $\pi(u) = u$ almost everywhere and π is one-Lipschitz.

For fixed n, the function $\sqrt{g_n}$ is continuous. Finite Fourier polynomials are uniformly dense in continuous complex functions on the torus, by the Stone–Weierstrass theorem applied to the self-conjugate character algebra. Choose p_j converging uniformly to $\sqrt{g_n}$. Then

$$\left\| |p_j|^2 - g_n \right\|_\infty \leq \|p_j - \sqrt{g_n}\|_\infty \left(\|p_j\|_\infty + \|\sqrt{g_n}\|_\infty \right) \longrightarrow 0.$$

Since m is a probability measure, this implies L^2 convergence. Choose one such square within $1/n$ of g_n in L^2. The chosen squares converge to u. □

Proposition 9.2. *Under the hypotheses of Theorem 1.1, positivity of all finite heat matrices is equivalent to $S_t\mathcal{C} \subseteq \mathcal{C}$ for every $t \geq 0$.*

Proof. In the forward direction use (15), then approximate $u \in \mathcal{C}$ by squares using Lemma 9.1. Boundedness of S_t gives convergence of their images, and closedness of $\mathcal{C}$ gives $S_t u \in \mathcal{C}$.

Conversely, apply physical positivity to every $|f|^2$. Its image is a continuous finite polynomial, so almost-everywhere nonnegativity implies pointwise nonnegativity. Indeed, a negative value of its real part would give a nonempty open set of negative values; Haar measure on the torus assigns positive measure to every nonempty open set. Its imaginary part similarly vanishes everywhere if it vanishes almost everywhere. Evaluating (15) at $x = 0$ and allowing arbitrary real c_i proves nonnegativity of every heat-matrix quadratic form. Evenness already supplies symmetry. □

Completion of the proof of Theorem 1.1. Lemma 2.1, Proposition 3.2, Proposition 5.1 and Proposition 9.2 give all five equivalences and the finite Gamma hierarchy. The semigroup properties and Proposition 6.1 give the analytic assertions. □

For real bounded u with $0 \leq u \leq 1$, positivity and $S_t 1 = 1$ also give $0 \leq S_t u \leq 1$ almost everywhere. Thus the word Markov, when used for these operators, refers to this order and constant-preservation property on the physical function space. No sample-path regularity or filtration is built by the proof.

10 The sharp active-coordinate threshold

Let $a_j \geq 0$ and $\alpha_j > 0$ for $j = 1, 2, 3$, and set

$$\psi(k) = \sum_{j=1}^{3} a_j |k_j|^{\alpha_j}, \qquad 0^\alpha = 0 \quad (\alpha > 0). \tag{16}$$

A coordinate is active when $a_j > 0$. The generator is defined by these rates, with no alternative Euclidean radial power substituted for their coordinatewise sum.

Theorem 10.1. *For the symbol* (16)*, every condition in Theorem 1.1 holds if and only if*

$$a_j > 0 \implies \alpha_j \leq 2 \quad (j = 1, 2, 3). \tag{17}$$

If an active exponent exceeds two, a nonnegative finite Fourier square has a negative evolved value on a set of positive Haar measure at some positive time.

Proof. We first prove the positive cases, including the endpoint. For $0 < \alpha < 2$ put

$$I_\alpha = \int_0^\infty (1 - \cos s) s^{-1-\alpha}\, ds.$$

This is finite and strictly positive. Near zero use $0 \leq 1 - \cos s \leq s^2/2$; near infinity use $1 - \cos s \leq 2$; positivity follows on an interval where the cosine is not one. Changing variables gives, also at $x = 0$,

$$|x|^\alpha = I_\alpha^{-1} \int_0^\infty (1 - \cos(sx)) s^{-1-\alpha}\, ds. \tag{18}$$

For a real sample x_i and coefficients summing to zero,

$$\sum_{i,l} c_i c_l (1 - \cos(s(x_l - x_i))) = -\left| \sum_i c_i e^{isx_i} \right|^2 \leq 0.$$

Every individual integrand in (18) is integrable, so finite summation and integration commute. This proves conditional negativity of $|x-y|^\alpha$. At $\alpha = 2$, expansion gives directly

$$\sum_{i,l} c_i c_l (x_l - x_i)^2 = -2\left(\sum_i c_i x_i\right)^2 \le 0.$$

Restrict to each integer coordinate, multiply by $a_j \ge 0$ and add. If $a_j = 0$ its summand is identically zero regardless of the positive exponent. Thus (17) implies conditional negativity of (16), and the main theorem supplies every asserted positivity.

For necessity, fix an active coordinate j and sample $0, e_j, 2e_j$ with coefficients $(1, -2, 1)$. Their sum is zero. Directly summing the six off-diagonal terms gives

$$\sum_{i,l=0}^{2} c_i c_l \psi((l-i)e_j) = 2a_j(2^{\alpha_j} - 4). \tag{19}$$

If $\alpha_j > 2$, this is strictly positive, contradicting conditional negativity.

The same witness gives a physical counterexample without any appeal to an abstract converse. Let

$$p(x) = 1 - 2e_{e_j}(x) + e_{2e_j}(x).$$

Then $|p|^2 = 6 - 8\cos(2\pi x_j) + 2\cos(4\pi x_j) \ge 0$, and

$$(S_t|p|^2)(0) = 6 - 8e^{-a_j t} + 2e^{-a_j 2^{\alpha_j} t}. \tag{20}$$

This vanishes at $t = 0$ and has derivative $-2a_j(2^{\alpha_j} - 4) < 0$ there. It is therefore negative for all sufficiently small $t > 0$. Each evolved polynomial is continuous, so its negative set contains an open neighborhood of zero and has positive Haar measure. □

This proof also shows exactly why a zero-weight exponent must not be tested. For example $(a_1, a_2, a_3) = (1, 0, 0)$ and $(\alpha_1, \alpha_2, \alpha_3) = (2, 4, 10)$ give the ordinary one-coordinate quadratic heat evolution, with stationary modes in the other two directions. Its positivity is unaffected by the nominal inactive exponents. The all-zero symbol gives the identity semigroup and is included as well.

The integral representation (18) is a classical stable-symbol argument. In the formal development, fractional covariance matrices provide the positive interior case; the human proof supplies an equivalent direct integral argument. No new formal declaration for that reorganization is claimed.

11 Finite-time increment kernels

Assume from now on that ψ satisfies Theorem 1.1. For each $t \ge 0$ define

$$q_t(k) = 1 - e^{-t\psi(k)}. \tag{21}$$

This symbol is normalized, even and nonnegative. It should be distinguished from the original generator symbol ψ.

Proposition 11.1. *The anchored kernel $C_t = K_{q_t}$ is positive semidefinite at every $t \geq 0$, and*

$$2C_t(k,l) = e^{-t\psi(l-k)} - e^{-t\psi(k)} - e^{-t\psi(l)} + 1. \tag{22}$$

If a torus-valued random variable X_t has Fourier law $\mathbb{E}e_k(X_t) = e^{-t\psi(k)}$, then

$$2C_t(k,l) = \mathbb{E}\big[\overline{(e_k(X_t) - 1)}(e_l(X_t) - 1)\big]. \tag{23}$$

Proof. For any zero-sum real c_i, the constant part of q_t cancels, so

$$\sum_{i,j} c_i c_j q_t(k_j - k_i) = -\sum_{i,j} c_i c_j e^{-t\psi(k_j - k_i)} \leq 0$$

by the finite heat-matrix positivity. Lemma 2.1 gives positivity of C_t. Expanding its definition proves (22). Expanding the product of character increments gives (23) under the stated Fourier law, using $e_{-k}e_l = e_{l-k}$ and evenness. □

The expectation in (23) is an uncentered second-moment Gram kernel of the increments $e_k(X_t) - 1$. Those increments need not have mean zero. The word covariance in "anchored covariance kernel" denotes the positive-kernel structure; it must not silently turn this identity into the centered covariance of those random variables. The frozen library constructs actual transition laws and fixed-step chains with this Fourier law. The proposition's matrix proof does not depend on a continuous-time path construction, and none is used here.

As $t \downarrow 0$, $q_t(k)/t \to \psi(k)$ for each frequency. Hence $C_t(k,l)/t \to K_\psi(k,l)$ on each finite sample. This is finite-dimensional entrywise convergence, not an operator-norm assertion over all frequencies.

12 Positivity does not imply increase in time

The scalar values $q_t(k)$ increase in t for every fixed k. Their anchored matrices need not increase in Loewner order. Recall that $C_t \succeq C_s$ means the difference is positive semidefinite on the chosen common finite sample.

Proposition 12.1. *Take $\psi(k) = k_1^2$, frequencies $k = e_1$ and $l = -2e_1$, and the real vector $b = (1, -2)$. For $s = \log 2$ and $t = 2\log 2$,*

$$b^{\mathsf{T}}(C_t - C_s)b = -\frac{16897}{131072} < 0. \tag{24}$$

Both C_s and C_t are positive semidefinite.

Proof. The rates of $k, l, l-k$ are $1, 4, 9$. Writing $r = e^{-u}$ gives

$$C_u(k,k) = 1 - r, \qquad C_u(l,l) = 1 - r^4, \qquad C_u(k,l) = \frac{1 - r - r^4 + r^9}{2}.$$

Consequently

$$b^{\mathsf{T}} C_u b = 3 + r - 2r^4 - 2r^9.$$

At $u = s$ one has $r = 1/2$; at $u = t$ one has $r = 1/4$. Subtracting gives

$$\left(\frac{1}{4} - \frac{1}{2}\right) - 2\left(\frac{1}{256} - \frac{1}{16}\right) - 2\left(\frac{1}{262144} - \frac{1}{512}\right) = -\frac{16897}{131072}.$$

The quadratic symbol meets (17), so each individual matrix is positive by Proposition 11.1. □

The example distinguishes two assertions: positivity at each time and positivity of time differences. It does not contradict a curvature theorem whose hypotheses and conclusion are different from Loewner monotonicity of C_t.

There is a similar distinction within the iterated Gamma hierarchy. For $\psi(k) = \frac{1}{2}|k_1|$ and $f = e_{e_1}$, (7) gives

$$\Gamma_n(f,f)(x) = 2^{-n}.$$

Thus $\Gamma_2(f,f) = 1/4 < 1/2 = \Gamma_1(f,f)$ despite positivity of every iterate. No inequality $\Gamma_2 \geq \Gamma_1$ follows from the hierarchy alone. The classical Bakry–Émery calculus [3] asks for specified quantitative comparisons, with their normalization and domain hypotheses; the jump-type power symbols here also need not satisfy the diffusion chain rule. The finite algebraic result must retain those distinctions.

13 Quadratic flatness on the finite core

For the same one-coordinate quadratic symbol, the precise pointwise curvature statement has sharp constant zero. This conclusion has its own proof; it is not inferred from Proposition 12.1.

Proposition 13.1. *For $\psi(k) = k_1^2$, one has $\Gamma_2(f,f)(x) \geq 0$ for every $f \in \mathcal{P}$ and every x. There is no $\rho > 0$ such that*

$$\Gamma_2(f,f)(x) \geq \rho\Gamma_1(f,f)(x) \tag{25}$$

for all real-valued $f \in \mathcal{P}$ and all $x \in \mathbb{T}^3$.

Proof. The first assertion follows from the main theorem, or directly from $K_\psi(k,l) = k_1 l_1$ in (8). To test (25), let

$$f = ie_{e_1} - ie_{-e_1} = -2\sin(2\pi x_1).$$

This polynomial is real-valued. In its two-mode presentation the frequencies are $e_1, -e_1$, the coefficients are $i, -i$, and at the origin the first kernel matrix is $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$. Its energy is $|i-(-i)|^2 = 4$. The entrywise square matrix is the all-ones matrix, whose energy is $|i+(-i)|^2 = 0$. Thus $\Gamma_1(f,f)(0) = 4$ and $\Gamma_2(f,f)(0) = 0$. No positive ρ can satisfy (25). □

For a check on normalization, set $D = (2\pi)^{-1}\partial_{x_1}$ on $\mathcal{P}$. Then $L_\psi = D^2$, $\Gamma_1(f,f) = |Df|^2$ and $\Gamma_2(f,f) = |D^2 f|^2$. These identities follow either by differentiation or from the same mode formula. They reproduce the sine values above. Positivity on the physical cone therefore coexists with sharp zero pointwise curvature in this example. The finite core is the domain of this statement; no extension to arbitrary members of $\mathrm{Dom}(A)$ is hidden in its notation.

14 Actual probability laws and fixed-step chains

The positive multipliers have an actual measure realization, not just an abstract coefficient interpretation. We retain the hypotheses of the main equivalence: $r : \Lambda \to [0,\infty)$ is even, $r(0) = 0$, and its anchored kernel is positive semidefinite. Write $P_t e_k = e^{-tr(k)} e_k$.

Theorem 14.1. *For each $t \geq 0$ there is a unique probability measure μ_t on $\mathbb{T}^3$ with*

$$\int e_{-k}(z)\, d\mu_t(z) = e^{-tr(k)} \qquad (k \in \Lambda).$$

These measures satisfy $\mu_0 = \delta_0$ and $\mu_s * \mu_t = \mu_{s+t}$, and depend weakly continuously on t. The translated laws $Q_t(x, A) = \mu_t(A - x)$ are Markov kernels satisfying Chapman–Kolmogorov. On continuous functions and on Haar-L^2 their integral operators agree with the original Fourier multipliers.

Proof. Let $B_N = \{-N, \dots, N\}^3$ and

$$q_N = |B_N|^{-1/2} \sum_{j \in B_N} e_j, \qquad h_N = |q_N|^2.$$

Orthogonality gives $\int h_N \, dm = 1$. By finite-square positivity, $g_{N,t} = P_t h_N$ is a continuous nonnegative polynomial of integral one. Thus $g_{N,t} m$ is a probability measure. Its kth Fourier moment is

$$e^{-tr(k)} a_N(k), \qquad a_N(k) = \frac{|\{j \in B_N : j + k \in B_N\}|}{|B_N|}.$$

For $M = \max_i |k_i|$ and $N \geq M$, the numerator lies between $|B_{N-M}|$ and $|B_N|$. Therefore $a_N(k) \to 1$ for every fixed k. Probability measures on the compact metric torus have a weakly convergent subsequence. Integrating each bounded continuous character along that subsequence gives all the required moments of its limit.

If two probability measures have these moments, their integrals agree on trigonometric polynomials. These form a uniformly dense subalgebra of $C(\mathbb{T}^3; \mathbb{C})$: they contain constants, are closed under conjugation and separate points. Uniform approximation and the bound $|\int f \, d\mu| \leq \|f\|_\infty$ extend equality to every continuous function. Continuous functions determine finite measures, proving uniqueness. At time zero δ_0 has the prescribed moments. Fubini and $e_k(x + y) = e_k(x) e_k(y)$ show that the moments of $\mu_s * \mu_t$ multiply to those of μ_{s+t}, so uniqueness proves the convolution identity.

If $t_j \to t$, every weakly convergent subsequence of μ_{t_j} has the moments of μ_t, by continuity of $e^{-tr(k)}$. Compactness and uniqueness force the full sequence to converge to μ_t. This proves weak continuity. For any Borel set A, $Q_t(x, A) = \int 1_A(x + z) \, d\mu_t(z)$ is Borel in x by measurability of addition and integration of a bounded Borel function. It has total mass one. Fubini and convolution give

$$\int Q_t(y, A) Q_s(x, dy) = Q_{s+t}(x, A).$$

Its action on e_k is $e^{-tr(k)} e_k(x)$, since evenness gives the same real moment at k and $-k$. The action on continuous functions is a contraction in the uniform norm. Moreover Jensen and Haar invariance give $\int |Q_t f|^2 dm \leq \int |f|^2 dm$. Density of polynomials in each space identifies these integral operators with the respective multiplier extensions. This proves the last assertion on the original physical Haar space, not just on a coefficient space. □

Fix a step $h \geq 0$. On the countable product probability space with independent coordinates Z_j of law μ_h, set $X_0 = x$ and $X_n = x + Z_1 + \cdots + Z_n$. This is a human reorganization of the source's Ionescu–Tulcea construction. Independence gives, first for bounded Borel f and then as a regular conditional law,

$$\mathbb{E}[f(X_{n+1}) \mid X_0, \dots, X_n] = Q_h f(X_n).$$

Indeed the finite past is measurable in $Z_1, \dots, Z_n$ and Z_{n+1} is independent of that larger sigma field; conditioning the resulting identity again gives the displayed formula. Convolution shows $\mathcal{L}(X_n) = Q_{nh}(x, \cdot)$, including $h = 0$ and $n = 0$. The finite-dimensional probabilities are the

iterated kernels $\delta_x(dx_0)\prod_{j=1}^n Q_h(x_{j-1},dx_j)$, which also identify the resulting countable-path law with the source trajectory law. No sample paths at non-grid times have been constructed here.

For $x=0$, the actual first-step character increment pairing is

$$\mathbb{E}[(e_{-k}(X_1)-1)(e_l(X_1)-1)] = e^{-hr(k-l)} - e^{-hr(k)} - e^{-hr(l)} + 1. \tag{26}$$

Expand the product and use its three character moments to prove this identity. Every integrand is bounded, so no interchange requires an extra domain. Its right derivative at $h=0$ is $r(k)+r(l)-r(k-l)=2K_r(k,l)$. Varying h varies the law of the first-step chain; this derivative is not a claim of differentiable sample paths or of an unsmoothed pointwise Gamma domain.

15 Poisson mixtures, an atom and a jump covariance

Fix an original heat step $\delta \geq 0$ and a finite Poisson parameter $\lambda \geq 0$. Define an actual probability measure by

$$\nu_\lambda = e^{-\lambda}\sum_{n=0}^{\infty}\frac{\lambda^n}{n!}\mu_{n\delta}. \tag{27}$$

At $\lambda=0$ use $0^0=1$ in the $n=0$ term. The weights are nonnegative and sum to one, so the sum defines a probability measure. This elementary Poisson subordination is classical; see Varadhan's primary notes, Sections 1.2 and 2.2 [1].

Theorem 15.1. *Put $q_\delta(k)=1-e^{-\delta r(k)}$. Then*

$$\widehat{\nu_\lambda}(k) = e^{-\lambda q_\delta(k)}, \qquad \nu_\lambda * \nu_\eta = \nu_{\lambda+\eta}, \qquad \nu_0=\delta_0.$$

The transformed symbol q_δ is normalized, even, nonnegative and has positive semidefinite anchored kernel. Its unique physical heat law at parameter λ is exactly ν_λ. Nevertheless

$$\nu_\lambda \geq e^{-\lambda}\delta_0, \qquad \nu_\lambda \not\ll m \quad \textit{for every finite } \lambda \geq 0. \tag{28}$$

Proof. Characters have modulus one, so the sum of their absolute integrals weighted by the Poisson probabilities is at most one. Thus integration commutes with the sum in (27), and

$$\widehat{\nu_\lambda}(k) = e^{-\lambda}\sum_{n\geq 0}\frac{(\lambda e^{-\delta r(k)})^n}{n!} = \exp[-\lambda(1-e^{-\delta r(k)})].$$

Multiplication of these moments and Fourier uniqueness prove the convolution assertion and the zero-parameter assertion. Normalization, evenness and nonnegativity of q_δ follow from those of r. Formula (26) for one original heat jump Z of law μ_δ gives

$$\mathbb{E}[(e_{-k}(Z)-1)(e_l(Z)-1)] = q_\delta(k)+q_\delta(l)-q_\delta(k-l) = 2K_{q_\delta}(k,l). \tag{29}$$

For finitely many coefficients c_j, multiplying by $\overline{c_i}c_j$ and summing gives the nonnegative integral $\mathbb{E}|\sum_j c_j(e_{k_j}(Z)-1)|^2$. Thus the anchored kernel is positive semidefinite. Theorem 14.1 applies to q_δ; its moments and uniqueness identify its law with (27).

The $n=0$ term in that sum is $e^{-\lambda}\mu_0 = e^{-\lambda}\delta_0$; every remaining term is a nonnegative measure. This proves the measure inequality, not merely positivity of one coefficient. Haar measure has null singletons: if a singleton had mass $a>0$, translation invariance would give the same mass to each of N distinct points, forcing $Na\leq 1$ for every N. Hence $m\{0\}=0$ while $\nu_\lambda\{0\}\geq e^{-\lambda}>0$. This proves failure of absolute continuity. The mass at zero may be larger than $e^{-\lambda}$; no equality is asserted. In particular $\delta=0$ yields $\nu_\lambda=\delta_0$ at all parameters. □

The source atom theorem assumes that the reference measure has a null origin singleton. The preceding Haar argument explicitly discharges that hypothesis here; it is a classical deduction, not an additional named Lean declaration. Thus actual positivity, conservativity and weak continuity coexist with failure of absolute-continuity smoothing.

There is also a bounded operator explanation. Let $U = P_\delta$ on $C(\mathbb{T}^3)$ or Haar-L^2. Since $\|U\| \le 1$, the series

$$V_\lambda = e^{-\lambda} \sum_{n \ge 0} \frac{\lambda^n}{n!} U^n = \exp[\lambda(U - I)]$$

converges in operator norm. The norm tail is bounded by the scalar exponential tail. The convolution identity gives $U^n = P_{n\delta}$, so V_λ is exactly integration against translated ν_λ, initially on bounded continuous functions and then on L^2 by contraction and density. If $A = U - I$, then $\|A\| \le 2$ and

$$\left\| \frac{e^{\lambda A} - I}{\lambda} - A \right\| \le \sum_{j \ge 2} \frac{\lambda^{j-1} \|A\|^j}{j!} \longrightarrow 0.$$

Consequently the generator in the parameter λ is the bounded operator $P_\delta - I$ on the whole space. This is an explicitly proved classical consequence, separate from the original unbounded generator domain for r. Bounded jump generation does not imply smoothing.

For each fixed λ, independent increments of law ν_λ give a countable-step chain exactly as above. Its first step from zero has the atom (28), and its character increment pairing is

$$e^{-\lambda q_\delta(k-l)} - e^{-\lambda q_\delta(k)} - e^{-\lambda q_\delta(l)} + 1.$$

The right derivative at parameter zero is $2K_{q_\delta}(k, l)$, which by (29) is the second moment of the increments of *one original heat jump*. It is an uncentered Gram pairing. A centered covariance would additionally subtract the product of the increment means. This changes the finite-parameter value, although that product has derivative zero at zero. The derivative varies the one-step Poisson parameter across a family of discrete chains; it does not construct a continuous-time compound-Poisson process with cadlag paths.

16 Formal source correspondence and editorial scope

The mathematical source is frozen at

`bab688bb6140cd6316c80680095a914ea0d346db.`

The namespace prefix is

`OperatorDerivativeRewrite.Foundation`

The accompanying `theorem-map.json` records full declaration names, exact lines, source hashes and baseline links. The main correspondences are as follows.

Paper statement	*Lean namespace and declaration*
Anchoring	`FourierCovarianceCriterion.symbol_iff`
Finite Gamma hierarchy	`FourierGammaCovariance.gamma_nonnegative`
Normalized first test	`FourierGammaNormalization.normalized_gamma_iff_negative_even`

Complex reality	`FourierGammaComplexCriterion.gamma_full_iff_covariance`
Finite heat criterion	`FiniteSchoenberg.heat_positive_iff_covariance`
Exact physical generator	`PhysicalTorusWaveSemigroup.generator_domain`
Closed graph	`PhysicalTorusWaveSemigroup.graph_closed`
Full cone closure	`PhysicalHeatPositiveCone.positive_l2_iff_square_closure`
Full cone preservation	`PhysicalHeatPositiveCone.physical_evolution_positive_all`
Mixed-symbol criterion	`AnisotropicMixedThreshold.anisotropic_negative_iff_active`
Physical sharp threshold	`PhysicalHeatPositiveCone.anisotropic_physical_positive_iff_active`
Finite-time kernel	`PhysicalHeatChain.heatIncrementSymbol_covariance`
Actual increment pairing	`PhysicalHeatChain.chainLaw_increment_pair_gamma`
Exact negative difference	`PhysicalHeatCovarianceNonmonotonicity.covariance_increment_energy_eq`
Actual physical laws	`PhysicalHeatFiniteBox.exists_physical_heat_law`
Poisson mixture and atom	`PhysicalHeatCompoundPoisson.compoundHeatLaw_character` and `PhysicalHeatCompoundPoissonAtom.noJump_dirac_le`
Sharp zero core curvature	`PhysicalHeatFlatCurvature.no_positive_quad_curvature`

The finite-polynomial graph-core assertion is a classical consequence of the exact domain, proved by truncation in Proposition 6.1; no separate three-dimensional physical graph-core declaration is claimed in the source map. The explicit square calculation also shows failure throughout a sufficiently small right neighborhood of zero, whereas the mapped physical counterexample declaration asks only for one positive time. These elementary deductions are proved here, not attributed to additional named Lean theorems. All physical statements use the unit-period Haar torus. The baseline receipt records a full build and 54,943 declarations audited with the standard three axioms; its evidence hashes were checked. No new Lean build or mathematical source change is claimed.

This paper is a coherent exposition of classical positivity structure and its verified physical realization, with explicit scope and counterexamples. Schoenberg's primary paper [2] is an antecedent for the exponential positive-definiteness equivalence, not merely a related formalization. The power threshold and integral argument are classical. We do not claim novelty for those statements or for Schur products. Priority for the exact finite witnesses and this combined formalization package remains unresolved. The Poisson mixture, bounded jump generator and no-jump mechanism are classical; Varadhan's primary notes were checked at the characteristic-function and operator-series formulas. No priority is claimed for them. The countable independent-increment realization reorganizes the source's kernel trajectory proof without asserting continuous-time path regularity.

The complete L^2 generator domain does not identify a closed unsmoothed pointwise Gamma domain. Gamma remains on the finite Fourier algebra. No continuous-time path regularity, positive curvature constant or physical gauge-field mass gap is established.

References

[1] S. R. S. Varadhan, *Poisson Processes*, Spring 2006 lecture notes, Sections 1.2 and 2.2. `https://math.nyu.edu/faculty/varadhan/spring06/spring06.1.pdf`.

[2] I. J. Schoenberg, Metric spaces and positive definite functions, *Trans. Amer. Math. Soc.* **44** (1938), 522–536. doi:10.1090/S0002-9947-1938-1501980-0.

[3] D. Bakry and M. Émery, Diffusions hypercontractives, *Séminaire de Probabilités XIX*, Lecture Notes in Math. **1123**, Springer, 1985, 177–206. `https://www.numdam.org/item/SPS_1985__19__177_0/`.